MILITARY SPECIFICATION

RACK, STORAGE, PIPE AND BAR STOCK, SINGLE AND DOUBLE FACE, ADJUSTABLE CANTILEVER ARMS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope.- This specification covers two column single and double cantilever armed open racks, for warehouse use (see 6.1).

1.2 Classification.- Racks shall be of the following types, sizes and capacity as specified (see 6.2).

Type I	-	Single face column with 6 arms
Size 1	-	24 inch long arms
Capacity A	-	2000 pound capacity at midpoint of arm
Capacity B	-	4000 pound capacity at midpoint of arm
Size 2	-	48 inch long arms
Capacity A	-	3500 pound capacity at midpoint of arm
Type II	-	Double face column with 12 arms
Size 1	-	24 inch long arms
Capacity A	-	2000 pound capacity at midpoint of arm
Capacity B	-	4000 pound capacity at midpoint of arm
Size 2	-	48 inch long arms
Capacity A	-	3500 pound capacity at midpoint of arm
Style A	-	36 inch spacing between column centerlines
Style B	-	48 inch spacing between column centerlines

FSC 7125

Source: https://assist.dla.mil -- Downloaded: 20260-21T22:19Z
Check the source to verify that this is the current version before use.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

TT-C-490	-	Cleaning Methods and Pretreatment of Ferrous Surfaces
PPP-C-650	-	Crate, Wood, Open and Covered

MILITARY

MIL-P-116	-	Preservation, Methods of

STANDARDS

FEDERAL

FED-STD-595	-	Colors

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	-	Marking for Shipment and Storage
MIL-STD-130	-	Identification Marking of U.S. Military Property

(Copies of specifications, standards and drawings required by supplier in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.- The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply:

National Motor Freight Traffic Association, Agent

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Associations, Inc., Tariff Order Section, 1616 P Street, N.W., Washington, DC 20036.)

Source: https://assist.dla.mil -- Downloaded: 202610-21T22:19Z
Check the source to verify that this is the current version before use.

Uniform Classification Committee, Agent

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 First article.- When specified (see 6.2), the supplier shall furnish a sample for first article inspection and approval (see 4.2, 6.2, and 6.3).

3.2 Materials and components.- The materials and components not definitely specified shall be of the quality normally used by the manufacturer for these type racks provided the complete item complies with all provisions of this specification.

3.3 Design and construction.- Rack components shall consist of steel columns, base, spacer bracing and adjustable arms (see Figure 1). Like components shall be identical and completely interchangeable. There shall be 6 arms per column for type I and 12 arms per column for type II racks. The unloaded arms shall have a tilt upward of no greater than 5/16 inch per foot of slope. The loaded arms shall support load requirements as specified in 1.2 and the assembled rack shall be capable of withstanding the tests specified in 4.4.1 and 4.4.2.

3.3.1 Columns.- Each column shall be a unit consisting of an upright and base support. A tapered upright may be provided at the option of the supplier. Columns shall have receptacles so that arms may be adjustable on 2- or 3-inch centers from 6 inches above base to 6 inches from top. The column and base must be so designed that if one side of the column is empty and the other is fully loaded, the rack will not fall over when the rack is not anchored to the floor. Each end of the base support shall have a 5/8 inch + 1/8 - 0, hole to permit anchoring to the floor.

3.3.2 Arms. Each end of the arm shall fasten securely to the column by use of a boltless arrangement. A safety catch feature shall be provided on each arm to securely lock the arm to the column at assembly. Design of the catch feature shall also be such that the arm cannot be dislodged by impact or movement of the rack (see 4.4.2). Components used for the catch shall not interfere with placement or withdrawal of the loads stored on the racks. The arms shall not deflect downward beyond horizontal when tested as specified in 4.4.1 using a safety factor of 1.65 which has been included in the loads indicated in 4.4.1.

Source: https://assist.dla.mil -- Downloaded: 20260-21T22:19Z
Check the source to verify that this is the current version before use.

3.3.3 Spacer bracing section.- Each spacer bracing section shall be a unit to accurately space the columns to the dimensions and configuration shown in figure 1. The frame may be attached without bolts provided the bracing gives stability to prevent "racking", "skewing" or "swaying" of a unit when all arms are capacity loaded. When specified (see 6.2), the frame assembly may be of the adjustable single frame design which can be used to provide either a 36-inch or 48-inch column spacing.

3.4 Leveling shims.- When specified (see 6.2), steel shims approximately 1/8" thick with a hole matching the hole in base support shall be furnished. The shims shall be of the same length and width as the base.

3.5 Finish.- All surfaces of the racks including hardware shall be cleaned and prepared for painting in accordance with any type or method of TT-C-490. All prepared surfaces shall be coated with at least one coat of enamel in accordance with the manufacturers standard commercial practice. The finish coat shall be gray, approximating color No. 16187 or 26173 of FED-STD-595.

3.6 Marking for identification.- Racks shall be marked for identification in accordance with MIL-STD-130.

3.7 Erection instructions.- Printed erection instructions shall be furnished with each rack unit.

3.8 Workmanship.- Welded joints shall be sound, smooth, free from pits, holes, fissures, and rough projecting edges; and shall be thoroughly fused to the base metal. Holes shall be accurately punched, drilled or reamed and adjacent edges shall have no burrs or fins to interfere with proper seating. All edges shall be free of burrs, slivers and projections. The enamel coat shall be smooth, evenly applied, free of sags, excess build-up, orange peel, runs, drips and shall have no uncoated areas.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 First article inspection.- When a first article sample is required, it shall be a completely fabricated rack and shall be examined for defects

Source: https://assist.dla.mil -- Downloaded: 202610-21T22:19Z
Check the source to verify that this is the currentversion before use.

in tables I and II, dimensions specified and tested as specified in 4.4.1 through 4.4.2. The presence of any visual defect, dimensions not within specified requirements or failure to pass the tests shall be cause for rejection of the first article.

4.3 Inspection.- Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

4.3.1 Component and material inspection.- In accordance with 4.1 above, component and materials shall be inspected and tested in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.3.2 Process inspection.- Examination shall be made during the manufacturing process to determine compliance with the preparation for painting requirements in 3.5. Whenever nonconformance is noted, correction shall be made to affected item and process.

4.3.3 End item inspection.- The lot shall be all racks of one type, size, capacity and style offered for inspection at one time. The sample unit shall be one complete knocked down rack ready for assembly.

4.3.3.1 Visual examination of knocked down rack.- Knocked down units shall be examined for defects in table I. The inspection level shall be II and the acceptable quality level (AQL) shall be 2.5 for major defects and 10.0 for total defects, expressed in terms of defects per hundred units.

TABLE I.- Visual examination

Examine	Defect	Classification Major	Minor
Finish	Type of surface finish or color not as specified	X	
	Poor adhesion, i.e., peeling, blistered or flaky	X	
	Area of no film	X	
	Tacky or soft runs or sags or drips	X	
	Foreign matter imbedded in finish		X
	Color shade varies between components		X
Design, construction and workmanship	Any component missing	X	
	Any component broken, fractured or cracked	X	
	Sharp edge, sliver or burrs	X	

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T22:19Z
Check the source to verify that this is the current version before use.

TABLE I.- Visual examination (cont'd)

Examine	Defect	Classification Major	Minor
Design, construction and workmanship (cont'd)	Any component malformed, buckled, warped, bent out of shape or damaged	X	
	Any hole or slot missing or in wrong location	X	
	Bracing incorrect configuration	X	
Identification marking	Missing		X

4.3.3.2 Examination of assembly.- Sample units from the inspection lot defined in 4.3.3 shall be assembled and examined for defects in table II. The inspection level shall be S-2 and the AQL shall be 4.0 expressed in terms of defects per hundred units.

TABLE II.- Examination during assembly

Examine	Defect
Erection	Components poorly fit, i.e. loose or requires unusual force or modification to assembly.
	Any clip, stud or fastener or safety lock inoperative or does not perform design function.
	Any component buckled or misaligned.
	Assembly misaligned.
Instructions	Missing, illegible, incorrect or incomplete.

4.3.3.3 Dimensional examination.- Inspection shall be made of the erected end item for compliance with dimensions specified. The inspection level shall be S-2 and the AQL shall be 6.5 expressed in defects per hundred units.

4.3.3.4 Testing of the end item.- When a first article sample is not required, the initial production unit shall be tested in accordance with all tests under 4.4 without evidence of failure.

4.3.3.5 Examination of preparation for delivery.- An examination shall be made to determine that packaging, packing, and marking as required by section 5

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T22:19Z
Check the source to verify that this is the current version before use.

are complied with. Defects shall be as indicated in table III. The sample unit shall be one shipping container fully prepared for delivery, except that it need not be sealed. Defects of closure shall be examined on shipping containers fully prepared for delivery. The lot shall be all containers offered for inspection at one time. The inspection level shall be S-2 and the AQL shall be 4.0 expressed in terms of defects per hundred units.

TABLE III.- Examination of preparation for delivery

Examine	Defect
Markings (exterior and interior containers)	Missing, incomplete, illegible, of improper size, location, sequence, or method of application.
Materials	Component missing or damaged.
Workmanship	Inadequate application of components such as incomplete closure of container flaps, loose strapping, inadequate stapling, bulging, or distortion of container.
Erection instructions	Missing.

4.4 Tests.-

4.4.1 Load test.- The assembled rack shall have its arms loaded at midpoint as follows: type I, size 1, capacity A - 3300 pounds; type I, size 1, capacity B - 6600 pounds (4 arms only); type I, size 2, capacity A - 5775 pounds. For type II racks, all arms on one side shall be loaded first to test for overturning before the complete rack is loaded. Loads shall be applied for one hour. Any evidence of slewing, overturning (see 3.3.1), deflection greater than that specified in 3.3.2, structural failure or permanent deformation after removal of load shall constitute failure of this test.

4.4.2 Safety catch.- A complete rack shall be assembled with the arms locked in place. The arms shall be tested for lateral, vertical and longitudinal movement by hand applied force of at least 15 pounds without coming apart from column. Any nonconformance to 3.3.2 shall constitute failure of this test.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T22:19Z
Check the source to verify that this is the current version before use.

5. PREPARATION FOR DELIVERY

5.1 Packaging.- Packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A.- Racks shall be disassembled and packaged knocked down. Like components of racks shall be bundled together and each bundle secured with waterproof tape or annealed copper wire. Erection instructions shall be packaged in accordance with Method IC-3 of MIL-P-116. Hardware removed during disassembly shall be packed in a cotton cloth drawstring bag and the bag secured to one of the components.

5.1.2 Level C.- Racks shall be packaged to afford adequate protection against damage during shipment from the supply source to the first receiving activity. The supplier may use his standard practice when it meets this requirement.

5.2 Packing.- Packing shall be level A, B or C as specified (see 6.2).

5.2.1 Level A.- Racks, packaged as specified in 5.1, shall be packed in a crate conforming to type II, style A of PPP-C-650. Each level A bundle shall be prevented from abraiding adjacent bundles by the use of cushioning material, the contents waterproofed, and the crate closed and strapped in accordance with the appendix of PPP-C-650.

5.2.2 Level B.- Racks, packaged as specified in 5.1, shall be packed in a crate conforming to type II, style B of PPP-C-650. Each level A bundle shall be prevented from abrading adjacent bundles by the use of cushioning material, and the crate closed and strapped in accordance with the appendix of PPP-C-650.

5.2.3 Level C.- Racks, packaged as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. Containers shall be in accordance with Uniform Freight Classification Rules or National Motor Freight Classification Rules, as applicable.

5.3 Marking.- In addition to any special marking required by the contract or order, shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use.- The 24-inch, 2000 pound rated, 6 arm racks are intended for storing pipe, bar stock and shapes. The 24-inch, 4000 pound rated, 4 arm racks may be used for vehicle racking of 4 jeeps by 4 point supports. The column capacity is rated at 14,000 pounds. The 48-inch arms,

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T22:19Z
Check the source to verify that this is the current version before use.

3500 pound rated, 6 arm racks are intended for storing sheet metal, plywood and similar material. The column capacity is rated at 28,000 pounds.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Rack type, size, capacity, and style required (see 1.2).
(c) When a first article is required (see 3.1, 4.2 and 6.3).
(d) When adjustable spacer bracing frame is required (see 3.3.3).
(e) Quantity of shims required (see 3.4).
(f) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.3 First article.- When a first article is required, it shall be inspected and approved under the appropriate provisions of ASPR 7-104.55. The first article should be a preproduction unit. The first article should consist of one complete unit. The contracting officer should include specific instructions in all procurement instruments regarding arrangements for inspection and approval of the first article.

Custodian:

Army - GL
Navy - SA
Air Force - 84

Preparing activity:

Army - GL

Project No. 7125-0083

Review activities:

Army - ME, SM

User activity:

Navy - MC, YD

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T22:19Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T22:19Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/ vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

S/N 0102-014-1802

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T22:19Z
Check the source to verify that this is the current version before use.

FOLD

POSTAGE AND FEES PAID



OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

Commander
US Army Natick Research and Development Command
ATTN: DRXNM-TD
Natick, MA 01760

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T22:19Z
Check the source to verify that this is the current version before use.